

Domain and range



1 For each of the following relations:

- i Find the domain.
- ii Find the range.
- iii Determine whether the relation is a function or not.

a $\{(8, 5), (1, 3), (3, 4), (9, 1), (2, 9)\}$

c $\{(4, 4), (8, 9), (2, 8), (7, 9), (3, 1)\}$

b $\{(2, 3), (3, 8), (6, 1), (8, 7), (3, 2)\}$

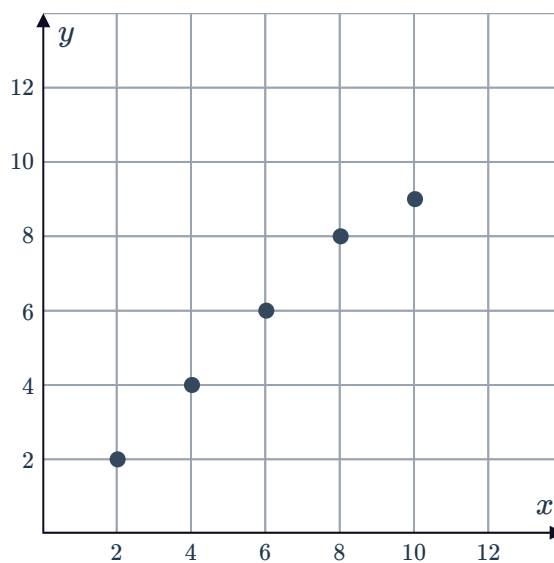
d

x	1	6	3	8	2
y	3	2	7	1	2

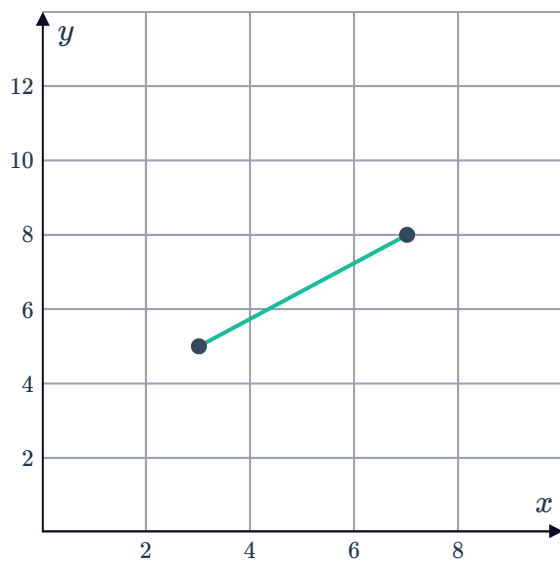
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x	7	7	8	5	3
y	1	9	3	2	6

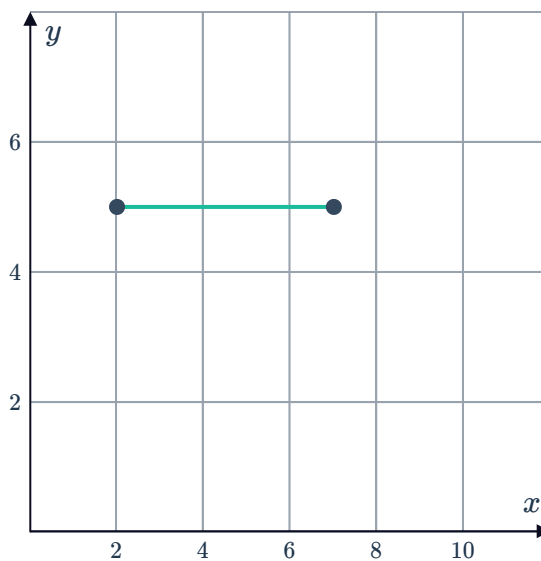
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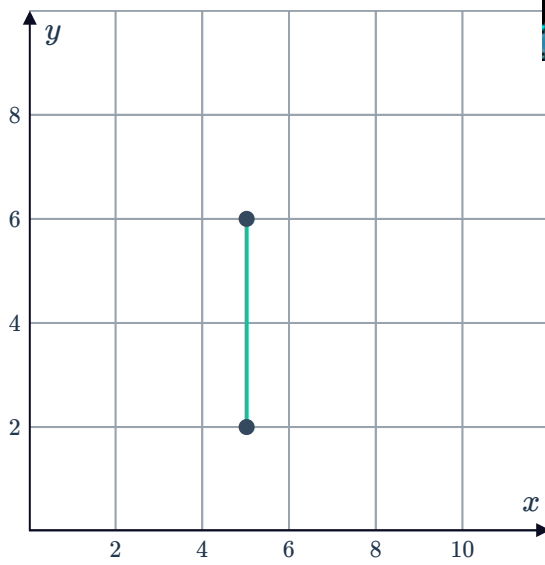
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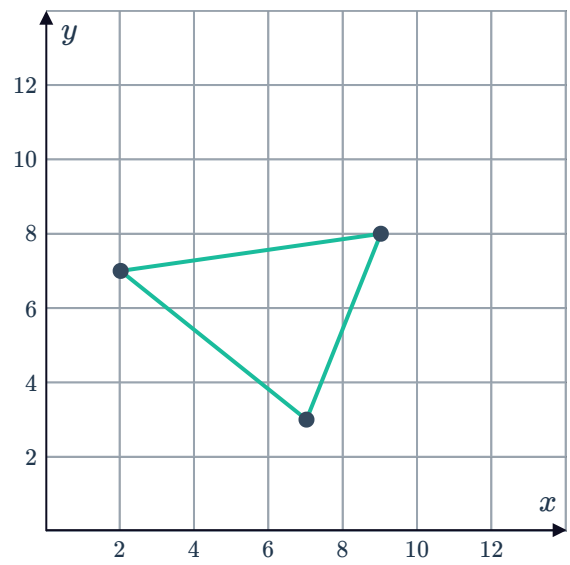
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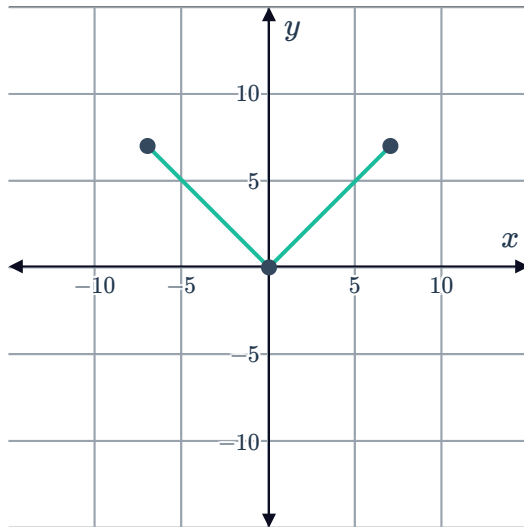
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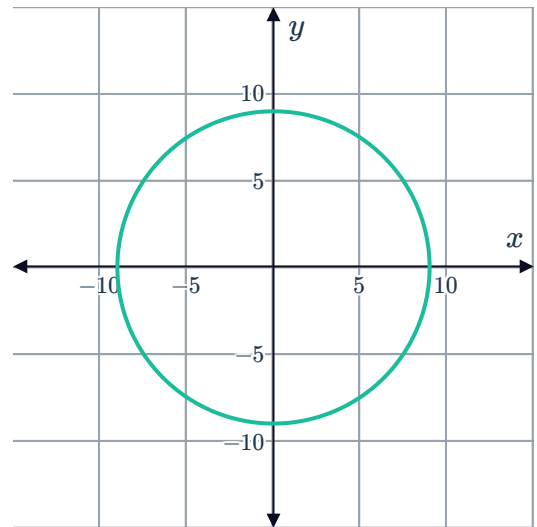
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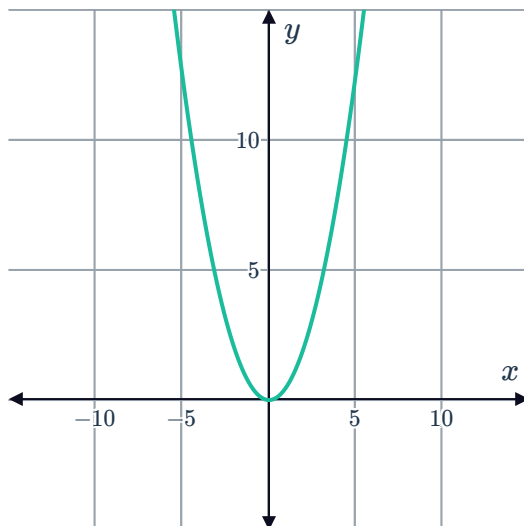
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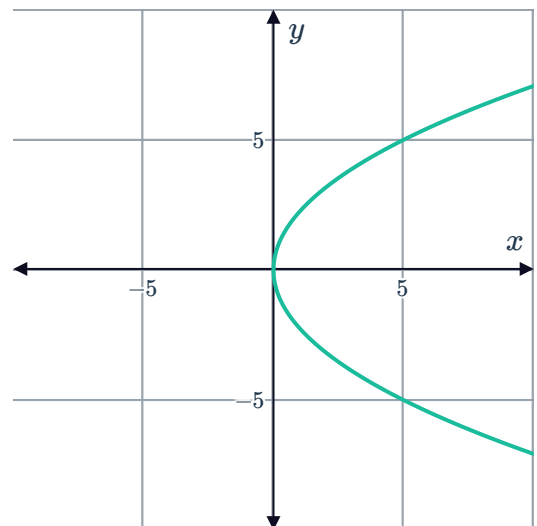
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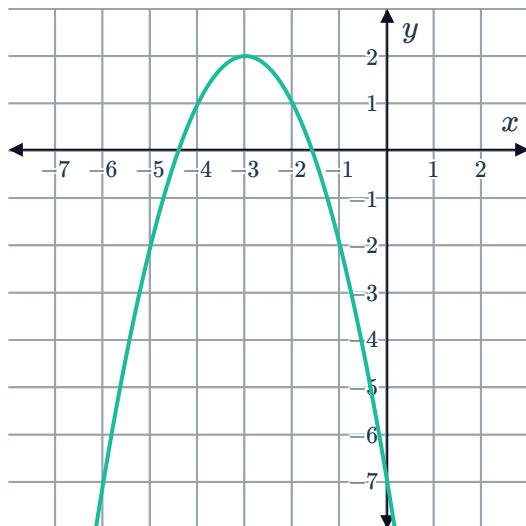
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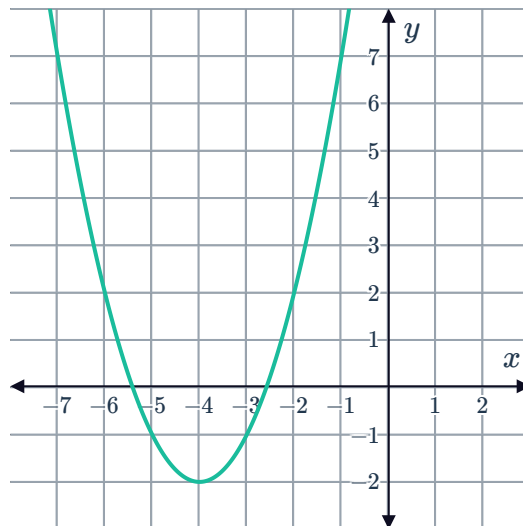
2 State the domain and range of the following functions:



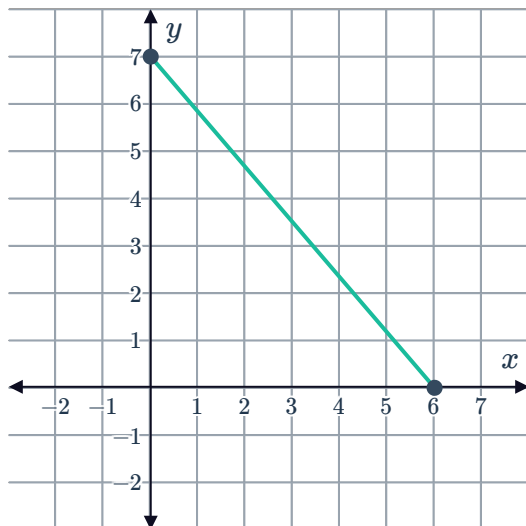
a



b



c



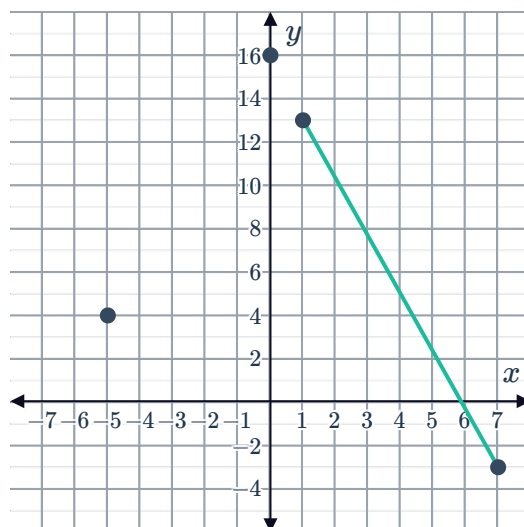
3 Consider the function f graphed. Use the graph to determine whether each of the given statements are true or false:

a The domain is $[-5, 7]$

b The range is $[-3, 16]$

c $f(1) - f(7) = 16$

d $f(0) = 16$



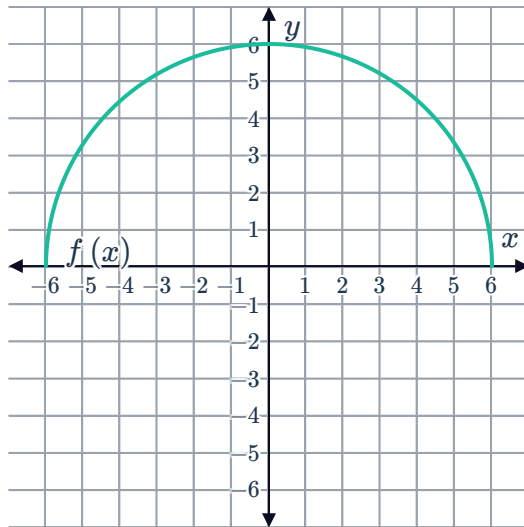
4 What is the domain of the following function?



a $f(x) = -3x - 1$

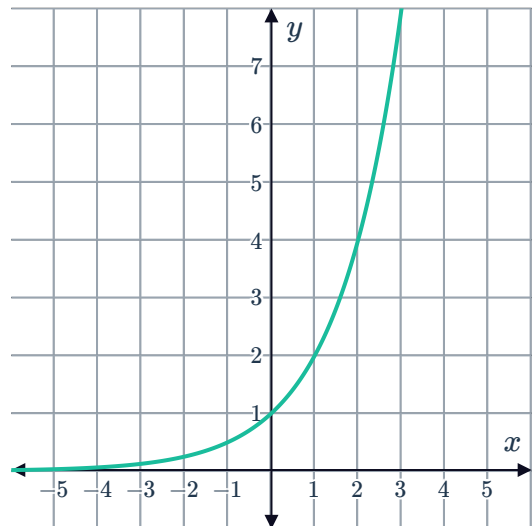
b $f(x) = 8x + 8$

c



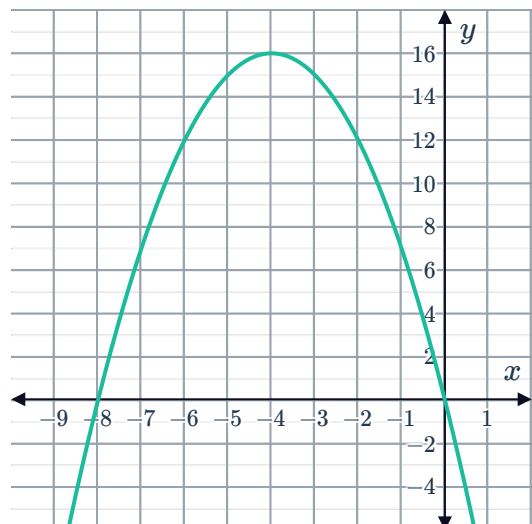
5 The graph of $y = 2^x$ is shown below:

- a What is the y -intercept of this graph?
- b Does the graph have an x -intercept?
- c What is the graph's domain?
- d What is the graph's range?
- e Find the value of y when $x = 7$.
- f Find the value of x when $y = 256$.



6 Consider the graph of the function shown:

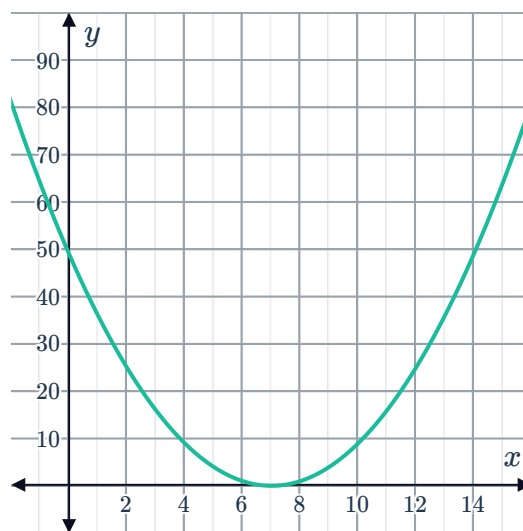
- a What is the maximum value of the graph?
- b Hence, determine the range of the function.
- c What is the domain of this function?



7 Consider the graph of the function shown



- a What is the minimum value of the graph?
- b Hence, determine the range of the function.
- c Over what interval of the domain is the function increasing?



8 For the following quadratic functions:

- i Determine whether the function has a maximum or a minimum point.
- ii Find the minimum or maximum value of the function.
- iii Find the range of the function.

a

x	y
1	17
2	7
3	1
4	-1
5	1
6	7
7	17

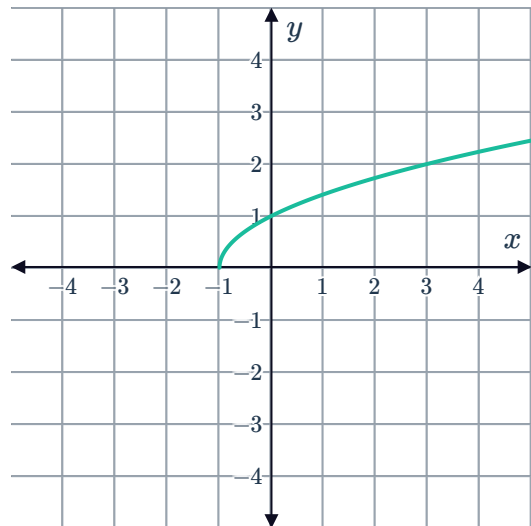
b

x	y
-5.5	-12
-5	-7
-4.5	-4
-4	-3
-3.5	-4
-3	-7
-2.5	-12

9 Consider the graph of the function $f(x) = \sqrt{x+1}$ below:



- a State the domain of the function.
- b Is there a value in the domain that can produce a function value of -2 ?



10 Consider the parabola defined by the equation $y = x^2 + 5$.

- a Is the parabola concave up or concave down?
- b What is the minimum y value of the parabola?
- c Hence, determine the range of the parabola.

Graphing functions

11 For the following functions:

- i Sketch the graph of the function.
- ii Find the domain.
- iii Find the range.

- a $f(x) = -x + 1$
- b $f(x) = 3x - 1$
- c $f(x) = -\frac{3}{2}x - 2$
- d $f(x) = -4$
- e $x = -1$

12 Consider the function $y = x^3 - 1$.

a Complete the table of values:

x	-2	-1	0	1	2
y					

- b Sketch the graph of the function.
- c Is there a value in the domain that can produce a function value of $3\frac{2}{3}$?

13 Consider $f(x) = x + 3$ for the domain $\{-5, -4, -3, -2, -1, 0, 1\}$.



a Complete the table of values:

b Plot the points on a number plane.

x	-5	-4	0	1
$f(x)$				

14 Consider $f(x) = -2x$ for the domain $\{-1, 0, 1, 2\}$.

a Complete the table of values:

b Plot the points on a number plane.

x	-1	0	1	2
$f(x)$				

15 Consider the function $f(x) = \frac{x+3}{2}$, for all real x .

a Complete the table of values:

b Sketch the graph of the function.

x	3	4	5	6	7
y					

16 Consider $-2x + y = 1$ for the domain $\{-4, -3, 0, 2\}$.

a Make y the subject of the equation.

b Complete the table of values:

x	-4	-3	0	2
y				

c Plot the points on a number plane.

17 Consider $f(x) = x^2 + 4$ for the domain $\{-3, -1, 0, 1, 3\}$.

a Complete the table of values:

b Plot the points on a number plane.

x	-3	-1	0	1	3
$f(x)$					

18 Consider $f(x) = (x + 2)^2$ for the domain $\{-2, -1, 0, 1, 2\}$.

a Complete the table of values:

x	-2	-1	0	1	2
$f(x)$					

b Plot the points on a number plane.

19 The function f is used to determine the area of a square given its side length.

a State whether the following values are part of the domain of the function:

i -8

ii 6.5

iii $9\frac{1}{3}$

iv $\sqrt{78}$

b For $n \geq 0$, state the area function for a side length of n .

c Sketch the graph of the function f .

Rational functions

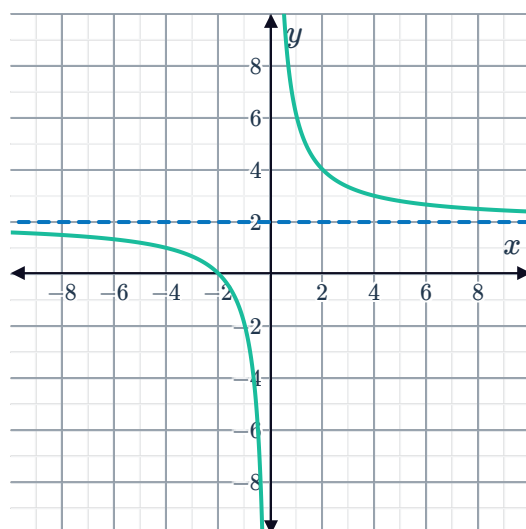
20 What is the domain of the function defined by $f(x) = \frac{1}{x+5}$?

21 For each of the following graphs of rational functions:

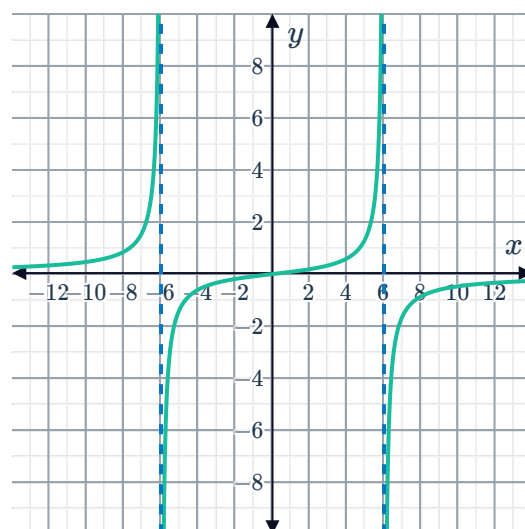
i State the domain using interval notation.

ii State the range using interval notation.

a



b





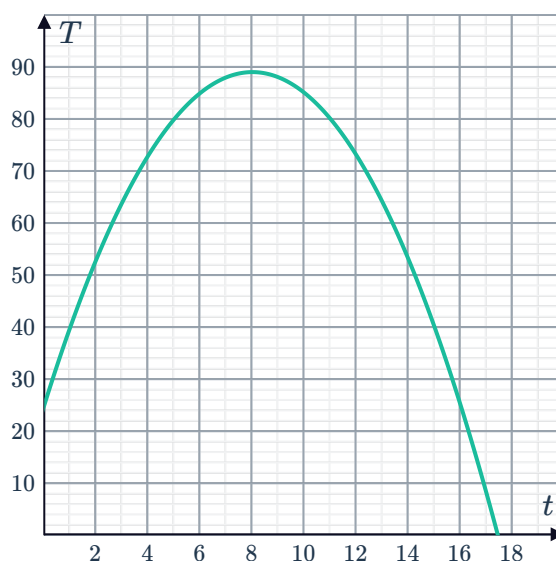
Applications

- 22** An industrial process involves heating a material from 25°C to 80°C in order to remove impurities.

Using the standard approach, the temperature of the material over time is given by the function:

$$T = -t(t - 16) + 25$$

A graph of the temperature is given, where temperature T is given in degrees Celsius ($^{\circ}\text{C}$) and time t is in minutes.



- a** What is the domain of this function?
- b** What is the range of this function?
- c** A new heating process is being developed that uses more energy but takes less time. This regime is based on a different quadratic function, and the table below shows the theoretical value of the temperature at different points in time using the new function.

t	0	1	3	4	30	31	32	33	34	35
T	25	45	80	96	96	80	63	45	25	4

For what values of t does this new function have the same range as the standard process?

- d** Hence, what values of t should be used for the new heating process?

- 23** The number of bees in a colony is initially measured and left to form a hive. The number of bees in the hive is measured each day over the course of one week. The function

$$H(x) = 30(3)^x$$

is found to model the number of bees, H , after x days.

- a** What is the domain of the function?
- b** What is the initial number of bees in the colony?
- c** Can the function be used to model the number of bees over an entire season?

24 At an indoor ski facility, the temperature is -8°C at the opening time of 9 am. At 10 am, the temperature is immediately brought down to -15°C and left for 3 hours before immediately taking it down again to -23°C where it stays until the close time of 7 pm.

- a** Write the piecewise function that models the indoor temperature $f(x)$ in terms of the number of hours after the facility has opened, x .
- b** The graph of the function is given. State the domain of the function.
- c** Kate entered the ski facility at 2:30 pm. What was the temperature inside the facility?
- d** Vincent wants to wait until the indoor temperature is -10°C or lower. When is the earliest he can enter the facility?

